

The resting membrane potential

The written notes for the resting membrane potential walkthrough. Week 4 competencies 1 to 5.

To print: these notes as a two-column PDF.

How to use these notes. Work through the walkthrough first, writing each prediction on your worksheet before you press Show me. Then read these notes to fill in anything you missed, and use them to review. They are the same facts in a form you can study from, not a copy of the slides.

Where the ions are

Ion	Inside (mM)	Outside (mM)
K ⁺	150	5
Na ⁺	15	145
Cl ⁻	10	108
Ca ²⁺	0.0001	1
Large anions (proteins, ATP)	Many	Almost none

- Textbooks differ a little in the numbers, but the pattern is the same.
- Potassium is the only one of these ions crowded inside. Sodium, chloride and calcium are crowded outside.
- The large anions are too big to cross the membrane. They are the negative charge left behind when potassium leaves.
- Ions cannot cross the lipid membrane on their own. They need channels, which are proteins built from genes.

How the inside becomes negative

- The cell has many potassium leak channels, which are always open.
 - Potassium is crowded inside, so it diffuses out through them.
 - Each potassium carries a positive charge away and leaves an unmatched negative charge behind, so the inside becomes negative. The charge gathers in a thin layer at the membrane, and only a tiny fraction of the potassium has to leave.
 - The negative inside pulls potassium back in. The pull grows as more potassium leaves.
 - When the pull in equals the push out, potassium moves in and out equally, with no net movement. That voltage is potassium's equilibrium potential, about -90 mV.
 - A few sodium leak channels let a little sodium in, carrying positive charge in and making the inside a bit less negative. Rest settles at about -70 mV.
- The ion with the most open channels has the most say over the voltage. At rest a neuron is roughly 25 times more permeable to potassium than to sodium.

- If the potassium leak channels were blocked, the inside would become less negative. The potassium leak is the main reason the inside is negative at all.

Resting membrane potential

The voltage across the membrane of a cell at rest, inside compared with outside, about -70 mV in a neuron.

Concentration gradient

The push on an ion from crowding. It ignores charge.

Electrical gradient

The push on an ion from charge. It ignores crowding.

Electrochemical gradient

The two pushes added together.

Equilibrium potential

The voltage at which the electrical push on an ion exactly balances its concentration push, so it has no net movement. Also called its balance point.

The two pushes on each ion

Ion	Concentration push	Electrical push at -70 mV	Result
K ⁺	Out	In	They oppose; a small net push out
Na ⁺	In	In	Both point in; sodium is pushed in hard

Ion	Equilibrium potential (about)
K ⁺	-90 mV
Na ⁺	+60 mV
Cl ⁻	-65 mV
Ca ²⁺	+120 mV

- At -70 mV, potassium is 20 mV from its equilibrium potential and sodium is 130 mV from its. That is why sodium's push is so much stronger.

The rule for which way an ion moves

- If an ion's channel is open, the ion moves toward its own equilibrium potential and drags the membrane with it.
- If the channel is closed, nothing moves, however strong the push.
- At rest, open sodium channels let sodium in and the membrane moves up toward +60 mV.
- At +30 mV, potassium is 120 mV from its equilibrium potential, so it rushes out and pulls the membrane back down. This is how a cell recovers after an action potential.

The Nernst and Goldman equations, in plain words

- You read these equations; you do not calculate with them in this course.
- The Nernst equation turns one ion's concentration difference into its equilibrium potential. Potassium at 150 mM inside and 5 mM outside gives about -90 mV.
- If outside potassium rises from 5 to 10 mM, potassium's equilibrium potential becomes less negative, about -72 mV, because the push out is weaker.
- The Goldman equation is the tug of war written as math: a weighted average of the equilibrium potentials, where each ion's weight is how permeable the membrane is to it.
- Potassium's weight is biggest, so rest sits near -90 mV; sodium's small weight pulls it to about -70 mV.
- More open potassium channels move rest closer to -90 mV (hyperpolarization). Many open sodium channels swing the membrane toward $+60$ mV, which is how an action potential begins.

The sodium-potassium pump

- The Na^+/K^+ ATPase uses ATP to move 3 Na^+ out and 2 K^+ in each cycle, carrying both ions against their gradients.
- Its main job is to keep the concentration gradients in place over minutes to hours, because the leaks never stop.
- Because it moves 3 positive charges out for every 2 in, it adds a few millivolts of negative charge directly.
- It does not set -70 mV from moment to moment; the leak channels do that. If the pump stops, the voltage barely changes at first, then drifts toward 0 mV over hours as the gradients run down. The pump is not the battery. It keeps the battery charged.

Words for changes in membrane potential

Depolarization

The membrane becomes less negative than rest (moves toward zero or above).

Overshoot

The part of an action potential where the inside is positive.

Repolarization

The membrane moves back toward its resting value, even while the inside is still positive.

Hyperpolarization

The membrane becomes more negative than rest.

- The words describe which way the voltage moved compared with rest, not whether the inside is positive or negative.
- Channels come in four kinds by what opens them: leak (always open), voltage-gated, ligand-gated (chemically gated) and mechanically gated. The channel gating walkthrough builds these in detail.

Excitability and blood potassium

- A neuron fires when its membrane reaches threshold, about -55 mV. The gap between rest and threshold is how big a push it needs. That is excitability.

- A cell closer to threshold is usually more excitable. If a cell stays depolarized for a long time, its sodium channels stay inactivated and it becomes less excitable.

Condition	Blood K^+ (normal 3.5 to 5.0 mM)	K^+ equilibrium potential	Resting potential	Effect
Hyperkalemia (missed dialysis)	7.0 mM	About -81 mV	About -65 mV, depolarized	At first closer to threshold; if it lasts, sodium channels stay inactivated and cells barely fire. In the heart: slow, disorganized conduction and risk of cardiac arrest.
Hypokalemia (loop diuretic)	2.8 mM	About -105 mV	About -76 mV, hyperpolarized	Farther from threshold, so muscles are weak; cramps. Low potassium also makes digoxin toxicity more likely.

- Treating hyperkalemia: IV calcium gluconate protects the heart by raising the threshold of heart cells, without changing potassium. Insulin with glucose moves potassium into cells by speeding the pump, which lasts only hours. Dialysis is the only one of the three that removes the potassium.
- Lidocaine blocks voltage-gated sodium channels, not the leak channels, so the resting potential barely changes, but no pain signal can be sent. A normal resting potential is necessary for a nerve to work, but it is not enough on its own.